5635131

B.Tech. DEGREE EXAMINATION, DECEMBER 2023.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORKS

(2013-14 Regulation)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

1. Write the purpose of Network Software?

2. What is the difference between Wired and Wireless Media?

3. Why Error Detection and Correction were necessary in network?

4. What is the usage of Bridges in networking?

5. Define Load Shedding.

6. What are the two types of IPv Addresses?

7. Define Sockets and its uses.

8. How Bandwith Allocation helps in sending packets?

9. What is Dynamic Web Page?

10. What is meant by DNS?

PART B — ($5 \times 11 = 55$ marks)

Answer ALL the questions.

UNIT I

11. Explain about Transmission Media and its types.

Or

12. Write in detail about Communication Satellites.

UNIT II

13. Write short note on:

(a) Error Control

(b) Flow Control.

Or

14. Describe in detail about:

(a) Routers

(b) Gateways

(c) Bridges

UNIT III

15. Explain in detail Distance Vector Routing examples.

Or

16. Write short note on:

(a) IPv IV

(b) IPv VI

UNIT IV

17. Describe in detail about Congestion Control

Or

18. Explain in detail architecture about RPC.

UNIT V

19. Explain in detail about Static and Dynamic Page.

Or

20. Write a short note on:

(a) HTTP

(b) Network Security.

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B.Tech. DEGREE EXAMINATION, MARCH 2021.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORKS

Time : Three hours Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions

All questions carry equal marks

1. Why are the networks organized as a stack of layers?

2. What is soft handover?

3. Mention any four error correcting codes.

4. What is piggybacking?

5. How do the adaptive algorithms differ from non adaptive algorithms?

6. How is ECN better than choke packets?

7. If the transport layer service is so similar to the network layer service, Why one layer not adequate?

8. The maximum payload of a TCP segment is 65,495 bytes. Why was such a strange number chosen?

9. When web pages are sent out, they are prefixed by MIME header. Why?

10. List any two substitution cipher algorithms.

PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each unit.

All questions carry equal marks

UNIT I

11. Describe the OSI Reference model. How does it differ from TCP/IP reference model?

Or

12. How are the communication satellites used for data transmission? Discuss its types.

UNIT II

13. Describe the Sliding Window Protocols.

Or

14. Why is Spanning tree used in connecting bridges? Discuss with example.

UNIT III

15. Describe the following.

(a) Broadcast Routing

(b) Multicast Routing

Or

16. Explain the congestion control approaches.

UNIT IV

17. How are connections established and rele
transport protocols? Explain.

Or

18. Describe UDP. Differentiate it with TCP.

UNIT V

19. Describe the e-mail architecture and servi

Or

20. Discuss the following.

(a) RSA Algorithm

(b) Transposition ciphers

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(c) What is the total length of the user datagram?

(d) What is the length of the data?

(e) Is the packet direct from a client to a server or vice versa?

(f) Draw the UDP header with length.

Or

8. Discuss the various congestion control mechanisms of TCP.

UNIT V

9. Encrypt the message "Have a Great Day" using Hill cipher with a key $\begin{pmatrix} 9 & 4 \\ 5 & 7 \end{pmatrix}$.

Or

0. Discuss DNS and explain the major section of DNS.

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B.Tech. DEGREE EXAMINATION, JANUARY 2022.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORKS

Time : Three hours Maximum : 75 marks

PART A – (10 × 2 = 20 marks)

Answer ALL questions.

1. List the Characteristics of Data Communication that determines the effectiveness of it.

2. Compare guided and unguided media.

3. Find the length of Burst error in this transmission a data sent is 0101101010001001 and the data received is 0110101010110111.

4. Define Flow Control.

5. Define Flooding.

6. Derive IP address and specify the IP class for the given binary notation .

   (a) 10011101 10001111 11111100 11001111

   (b) 11011101 10001111 10111111 11001111

7. Differentiate between UDP and TCP.

8. Define Multiplexing and demultipexing.

9. Recognize the duties of application layer and list any 2 protocols used in application layer.

10. Distinguish between public-key private–key cryptography.

PART B – (5 × 11 = 55 marks)

Answer ALL questions.

UNIT I

11. Sketch the OSI model and explain the functionalities of each layer in data communication.

Or

12. Illustrate the working of twisted pair and coaxial cable that could be used for transmission.

UNIT II

13. Discuss the method of error correction using Hamming code.

Or

14. Describe the functions of repeaters, hubs, bridges and gateways in a network

UNIT III

15. (a) What is the class of each of the following address?

(i) 10011101 10001111 11111 11001111

(ii) 221.143.253.15

(b) Write each of the following in dotted-decimal notation.

(i) 4.23.145.90

(ii) 227.34.78.7

(c) Find the net-id and host-id for each address

(i) 227.34.78.7

(ii) 126.23.145.90

(iii) 192.34.78.7

Or

16. Define routing and explain the link sate routing routing packets from source to destination with example.

UNIT IV

17. The dump of a UDP header in hexadecimal format is "CB84000D001C001C". Solve the following.

(a) What is the source port number?

(b) What is the destination port number?

UNIT V

19. Explain in detail about

(a) SMTP (6)

(b) HTTP (5)

Or

20. Draw and explain with a neat architecture diagram of SNMP and explain about the various services provided by SNMP.

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B.Tech. DEGREE EXAMINATION, SEPTEMBER 2020.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORKS

(2013-14 onwards)

Time : Three hours | Maximum : 75 marks

PART A — ($10 \times 2 = 20$ marks)

Answer ALL questions.

All questions carry equal marks.

1. Mention the components of a computer network.

2. Give examples of network topologies.

3. Distinguish between guided and unguided media.

4. What do you mean by multiplexing?

5. Expand CSMA and CDMA.

6. What are the variants of Ethernet?

7. Write notes on classful addressing of IP.

8. Define congestion.

9. What is meant by three-way handshake?

10. Write about DNS.

PART B — (5 × 11 = 55 marks)

Answer ALL questions ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Explain in detail about different guided and unguided media used in computer networks.

Or

12. With a neat diagram, briefly explain all the layers of OSI model and the services provided by each layer.

UNIT II

13. Write notes on CRC. How error detection can be achieved using CRC? Explain with an example.

Or

14. Write notes on :

(a) FDMA

(b) CDMA

(c) TDMA

UNIT III

15. Explain various IP addressing principles with suitable examples.

Or

16. What is ICMP? Explain ICMP header format and various ICMP messages with their applications.

UNIT IV

17. Explain in detail about various TCP congestion control mechanisms.

Or

18. With a neat diagram, give a detailed explanation about TCP state transition diagram.

UNIT – V

19. Recall the purpose of Domain Name System. Discuss the Hierarchy of Name Servers.

Or

20. Explain any three Traditional Symmetric-key Ciphers with example.

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B.Tech. DEGREE EXAMINATION, SEPTEMBER 2021.

Fifth Semester / Third Year

Computer Science and Engineering

COMPUTER NETWORKS

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Define a protocol.

2. State the suitable types of networks for a college, city and country.

3. List the advantages of gateways.

4. A signal carrying data in which two data element is encoded as one signal element if bit rate is 100 Kbps what is the average value of the baud rate if C is 0.5?

5. Find the network address for following IP address (a) 221.143.65.15 (b) 129.215.1.1

6. Recall the concept of pipelining in networking

7. Compare Go back-N and Selective Repeat ARQ.

8. Recall the services provided by TCP to the process at the application layer.

9. Identify the attacks threatening Integrity.

10. Recall the concept of Denial-of-Service.

PART B — (5 × 11 = 55 marks)

Answer ALL Questions, ONE from each Unit

All questions carry equal marks.

UNIT – I

11. Explain the OSI model with respect to the functionalities of each layer in data communication.

Or

12. Compare and contrast each type of network topology with their advantages and disadvantages.

UNIT – II

13. Discuss Error detection and error correction, which is difficult to identify, Justify with example.

Or

14. List and demonstrate various types of sliding window protocol with a neat diagram.

UNIT – III

15. What is an IP address? Justify the need for various classes of IP addressing.

Or

16. Define Routing and explain the distance vector routing of routing packets from source to destination with an example.

UNIT – IV

17. Recall the concept of Silly Window Syndrome. Also explain Nagle's algorithm and Clark's Solution.

Or

18. What is a 3-way handshake? Explain the three way handshake of establishing a TCP connection.

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B.Tech. DEGREE EXAMINATION, MAY 2019.

Fifth Semester

Computer Science and Engineering

COMPUTER NETWORKS

(2013-14 Onwards Regulations)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. What is the essential difference between computer networks and distributed systems?

2. Is the Nyquist theorem true for optical fiber or only for copper wire?

3. Data link protocols always put the CRC in a trailer rather than in a header, why?

4. The following character encoding is used in a data link protocol:

A:01000111; B:11100011 FLAG:01111110; ESC:11100000

Show the bit sequence transmitted (in binary) for the four character frame: A B ESC FLAG when each of the following framing methods are used.

5. Which of the following network devices uses layer 3 of the OSI model? Justify your answer

(a) Router

(b) Bridge

(c) Hub

(d) Switch.

6. Define tunneling.

7. Define piggy packing.

8. Define POP3.

9. Explain peep hole optimization.

10. What is PGP? Why do we use it.

20. (a) When web pages containing emails are sent out they are prefixed by MIME Header. Why?

(b) What is trivial file transfer protocol. Explain briefly?

(c) Why Gateways are used during mail transfer?

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PART B — (5 × 11 = 55 marks)

Answer ALL questions, ONE from each Unit.

All questions carry equal marks.

UNIT I

11. Ethernet and wireless networks have some similarities and some differences. One property of Ethernet is that only one frame at a time can be transmitted on an Ethernet. Do 802.11 share this property with Ethernet? Discuss your answer.

Or

12. Explain in detail about the OSI reference model of computer network architecture.

UNIT II

13. A link that connects two distinct computers. On the data link layer, the computers run the PAR protocol. Each data frame carries D bytes of payload. The round-trip propagation delay is R seconds. Assume that the link has an unrestricted bandwidth. Compute the throughput (in bytes per second) when no frames are lost on the link and when frames have no bit errors.

Or

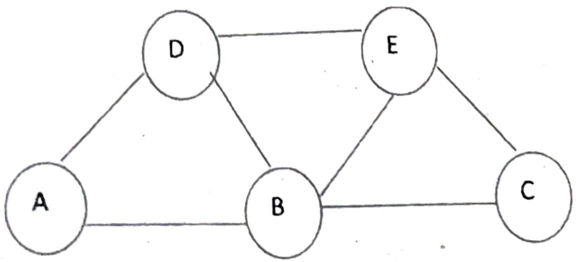
14. Explain the following terms
   - (a) Repeaters
   - (b) Bridges
   - (c) Routers
   - (d) Switches
   - (e) Gateways.

UNIT III

15. To address the shortage of IPv4 addresses, network Address Translation was developed
   - (a) Explain in detail the working of NAT
   - (b) Give four reasons why NAT violates some basic design principles of the internet.

Or

16. Compute the shortest paths from router A to any of the other routers in the figure by means of Dijkstra's shortest-path algorithm.



Edge Costs are

A to D:2, A to B:3, D to B:6, D to E:4,B to E.9, B to C : 8, E to C:7.

UNIT IV

17. (a) What are the main differences between
   and TCP/IP reference models?
   - (b) How congestion is controlled in TCP?

Or

18. (a) Is the TCP checksum necessary or could T
   allow IP to checksum the data.
   - (b) Describe the various characteristics of U
   protocol.
   - (c) Discuss the advantages of Electronics D
   Exchange (EDI).

UNIT V

19. (a) Write a short note on Network security.
   - (b) Illustrate the work of RSA algorithms.
   - (c) How optimization is achieved in DNS?

Or

14. Explain the following terms
   (a) Repeaters
   (b) Bridges
   (c) Routers
   (d) Switches
   (e) Gateways.

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15. To address the shortage of IPv4 addresses, network Address Translation was developed
   (a) Explain in detail the working of NAT
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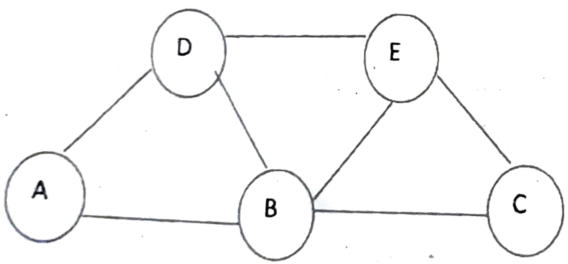

Edge Costs are

A to D:2, A to B:3, D to B:6, D to E:4,B to E.9, B to C : 8, E to C:7.

UNIT IV

17. (a) What are the main differences between O[illegible] and TCP/IP reference models?
   (b) How congestion is controlled in TCP?

Or

18. (a) Is the TCP checksum necessary or could T[illegible] allow IP to checksum the data.
   (b) Describe the various characteristics of U[illegible] protocol.
   (c) Discuss the advantages of Electronics D[illegible] Exchange (EDI).

UNIT V

19. (a) Write a short note on Network security.
   (b) Illustrate the work of RSA algorithms.
   (c) How optimization is achieved in DNS?

Or

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B.Tech. DEGREE EXAMINATION, NOVEMBER 2019.

Fifth Semester

CSE

COMPUTER NETWORKS

(2013-14 onwards)

Time : Three hours Maximum : 75 marks

PART A — (10 × 2 = 20 marks)

Answer ALL questions.

All questions carry equal marks.

1. Differentiate between connection - oriented and connectionless services.

2. Enumerate the issues of TCP/IP model and protocols.

3. Define the term piggybacking.

4. Write down the uses of Bridges.

5. Compare datagram and virtual-circuit networks

6. Give the format of IPV4 header.

7. Write the usage of transport service primitives.

8. Define – Error control and flow control.

9. Distinguish between static and dynamic web pages.

10. What does MIME and IMAP stands for?

PART B — (5 × 11 = 55 marks)

Answer ALL questions.

11. Illustrate the OSI reference model with different layers functionalities.

Or

12. Explicate the guided transmission media with necessary diagrams.

13. With suitable example, explain any two error-correcting and error-detecting codes.

Or

14. Narrate the various collision-free protocols.

15. Discuss in detail about the Link-State Routing.

Or

16. List and Explain the different approaches to prevent congestion.

17. Briefly describe the elements of transport protocols.

Or

18. Elucidate the connectionless transport protocol.

19. Illustrate the architecture and services of the email system.

Or

20. Elaborate the RSA public-key cryptography algorithm.